

Rate Coefficients of the Elementary Stages of Heterogeneous Catalytic Recombination of Dissociated Air on Thermal-Protective Coatings

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Abstract—On the basis of quantum-mechanical calculations within the framework of the cluster models and the literature data the rate coefficients of the elementary stages of the complete system of heterogeneous catalytic recombination of dissociated air on the surfaces of thermal-protective β -cristobalite and α -Al₂O₃ ceramic coatings are determined. The processes of impact and associative recombination of adsorbed oxygen and nitrogen atoms are taken into account.

Keywords: reusable thermal-protective coatings of spacecraft, heterogeneous catalytic reactions, cluster, quantum mechanics, density functional method.

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Great interest in an investigation of the interaction between multicomponent gas mixtures and low-catalytic materials is due to the development of reusable spacecraft and promising hypersonic flight vehicles. In high-heat regions of the atmospheric flight trajectory these vehicles are in nonequilibrium flow and the use of low-catalytic coatings reduces severalfold the heating at the expense of heterogeneous catalytic recombination of atoms. The considerable reduction of heat fluxes makes it possible to diminish the heat-shielding weight and to increase the payload [1–3].

In mathematically formulating this problem the catalytic activity of the surface enters in the boundary conditions for the system of equations governing multicomponent, chemically reacting flow past the surface. The heterogeneous catalytic recombination effects are usually described by introducing the recombination coefficients γ_i which show the fraction of the atoms which take part in recombination on the surface. The coefficients γ_i depend on both the surface properties and the gas phase conditions. These dependences include a great many parameters.

The very important quantities determining the temperature dependence of the coefficients γ_i are the rate coefficients of the elementary stages of heterogeneous catalytic recombination. At an appropriate choice of the rate coefficients of the elementary stages the kinetic models allowing for the detailed mechanism of heterogeneous catalytic reactions and based on Langmuir's theory of adsorbed layers [4] make it possible to satisfactorily describe the aerodynamic heating of the windward surface of reusable spacecraft along the entire re-entry trajectory in the Earth atmosphere [1, 3, 5, 6]. However, these coefficients include several parameters determined from the comparison with the experimental data on the heat fluxes; in the case of a multiparameter dependence this approach can be ambiguous [7].

The new approaches based on the methods of quantum mechanics and molecular dynamics allow one to better understand the mechanism of heterogeneous catalytic processes and to determine the rate coefficients of elementary processes without invoking the experimental data. In studies [8, 9] the coefficients of oxygen atom recombination and recombination energy accommodation on silicon thermal-protective coatings

(β -cristobalite and SiC) were calculated within the framework of classical molecular dynamics using the potentials of the interaction between oxygen and material atoms available in the literature.

In [10, 11] heterogenous recombination of oxygen atoms on an aluminum oxide surface was investigated using a semi-classical approach. In this approach the quantum mechanics methods, together with the density functional theory [12], were applied to determine the potential energy surface corresponding to the interactions of O(3 P) atoms with a cluster modeling the α -Al₂O₃ surface which was then used in molecular-dynamic calculations.

In this study, the interaction between dissociated air and thermal-protective β -cristobalite (111) and α -Al₂O₃ ceramic coatings is investigated on the basis of quantum-mechanical calculations. The rate coefficients of the complete system of the elementary stages of heterogenous catalytic recombination of dissociated air atoms on these materials are determined with account of adsorption/desorption and the impact and associative recombination of oxygen and nitrogen atoms.

1. MODELING OF THE MATERIAL STRUCTURE AND ADSORPTION PROCESSES

In dissociated air on materials under study the detailed mechanism of heterogenous catalytic processes presented below was taken into account and the rate coefficients of the corresponding elementary stages were calculated.

Adsorption/desorption of atoms



impact recombination of atoms (Eley–Rideal mechanism)



and associative atom recombination (Langmuire–Hinshelwood mechanism)



Since heterogenous catalytic reactions on surfaces involve hemosorbed particles, whose bonds with the surface are due to short-range forces, it is the cluster approach that is rather effective in describing local interactions. The cluster is a comparatively small fragment of the solid body lattice composed of a finite number of atoms of the surface layer and near-surface layers. Investigating the interactions with the cluster opens up possibilities for the direct use in calculations of quantum mechanics methods dealing with molecular systems of restricted dimensions.

1. The crystal α -Al₂O₃ surface was modeled by an Al₄O₆ cluster reproducing the crystal stoichiometry and the valent state of the Al atoms lying on the surface (Fig. 1). To model the adsorption of the oxygen and nitrogen atoms in the main state (O(3 P), N(4 S)) on the α -Al₂O₃ surface the oxygen atoms of the lower surface of the material were fixed in their positions occupied in the Al₄O₆ cluster. The positions of the other atoms were optimized at given coordinates R , θ , and φ of the adsorbed atom (Fig. 1). This model allows for surface monolayer relaxation due to the interaction between the adsorbent and the adsorbate. The calculations show the importance of the processes of the structural relaxation of surface atoms. In the calculations we used the B3LYP functional [13, 14] with the valent-split 6-31G* Pople basis [15]; this choice was confirmed by the comparison with the experimental data from the earlier authors' studies [10, 11].

2. β -cristobalite surface was modeled by the Si-terminated edge (111) of the β -cristobalite under the assumption of the edge-centered symmetry lattice $Fd3m$ for its ideal crystal, as shown in Fig. 2. It consists of equilateral triangles united in a hexagon with other Si atoms located on the outer layer of the surface, while the Si atoms of the second layer occupy the centers of each triangle. To describe adsorption of an outer atom

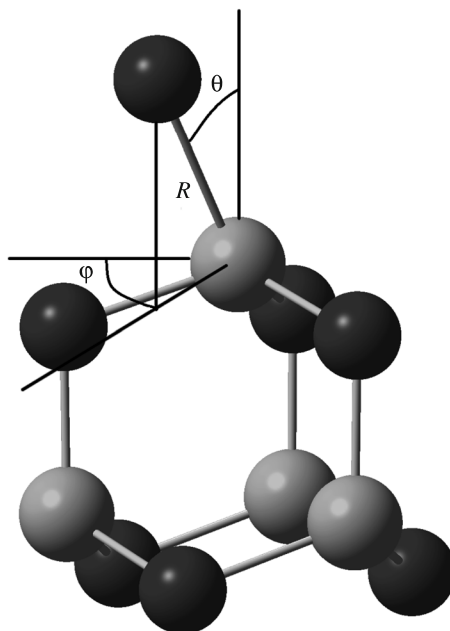


Fig. 1. Cluster model of oxygen atom adsorption on the Al_2O_3 surface; I are the Al atoms and II are the O atoms.

we take bridge Si–O–Si structures with Si atoms of the second layer and three bridges to the second-layer silicon with two Si–O bonds per atom. The structure thus obtained is shown in Fig. 3. It consists of 10 Si atoms, 12 O atoms, and 15 H atoms which are used for saturating the free surfaces of the cluster. The cluster possesses the point symmetry C_{3v} and one non-paired electron (total spin of the cluster $S_c = 1/2$).

All the calculations were performed within the framework of the method of the density functional theory with the functional and the basis described below. For an isolated $\text{Si}_{10}\text{O}_{12}\text{H}_{15}$ cluster we considered two configurations, namely, the “crystalline” one optimized with the constraint of equal lengths of the Si–O bonds and the “surface” one, in which for inner (Si, O, Si, O) planes were frozen. The second Si surface and the upper silicon atoms could move along the z direction perpendicular to the surface, while the oxygen atoms belonging to the first plane were able to consistently move in the z direction and in the vertical planes of symmetry of the C_{3v} group. The fixed parameters were taken on the basis of crystalline cluster optimization. The lengths and directions of the artificial Si–H bonds were optimized for the crystalline cluster and frozen in all last calculations. In the process of adsorption an atom from the gas phase approached the vertex (Si atom) of the z direction. The cluster optimization in the process of adsorption was made under the same constraints as for the surface cluster.

The tests of the accuracy and performance of different gradient-corrected and hybrid functionals were made on diatomic molecules pertinent to the problem discussed. The O_2 , N_2 , and NO molecules are the heterogeneous recombination products, while the SiO molecule qualitatively characterizes the description of the material structure. A preliminary analysis showed that in using the basis of the same or higher quality compared with 6-31G* the results much more strongly depend on the choice of the functional than on the further extension of the basis. Moreover, the calculations showed that the B3LYP functional ensures the greatest mean accuracy on the molecule and atom set under consideration.

2. CALCULATION OF THE RATE COEFFICIENTS OF THE ELEMENTARY STAGES

The quantum mechanical calculations gave the necessary information (activation energies and oscillation frequencies) for using the transitional state theory adapted to the surface reactions [16] in determining the coefficients of the elementary stages in the Arrhenius form.

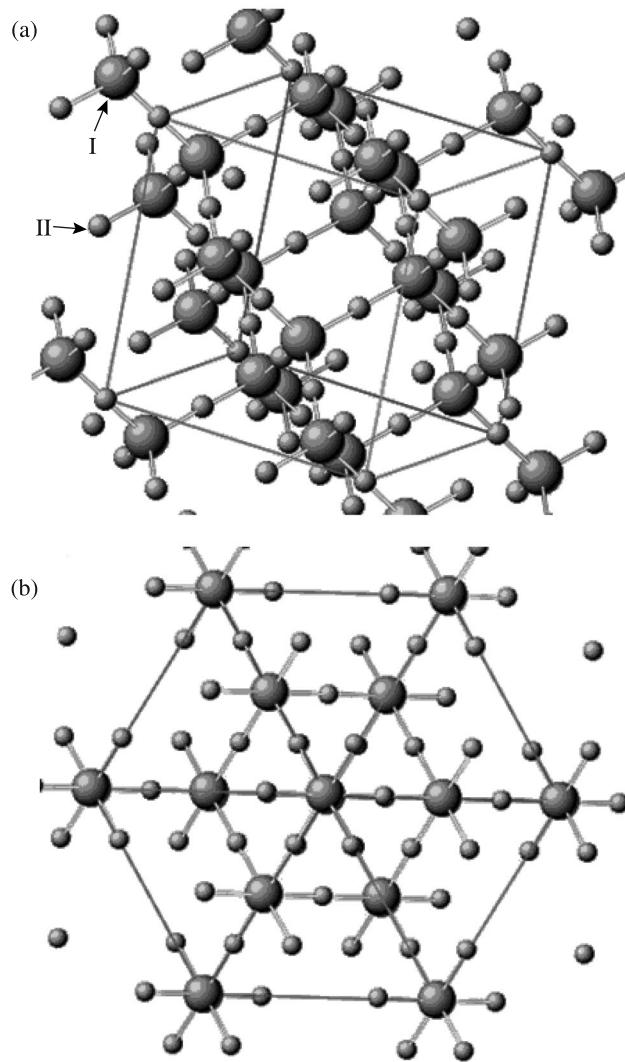


Fig. 2. Cell of an ideal crystal of β -cristobalite of the $Fd3m$ symmetry (a) and side (111) (b); I are the Si atoms and II are the O atoms.

Hemosorption/Desorption

The hemosorption rate coefficient can be represented in the form:

$$k_a = \frac{s_0(T)}{S^0} \left(\frac{k_B T}{2\pi m} \right)^{1/2},$$

where S^0 is the number of possible adsorption sites on the surface in the units of the inverse area, s_0 is the adhesion coefficient, k_B is the Boltzmann constant, and m is the adsorbed atom mass.

The parameter S^0 was calculated from the geometric considerations for a single adsorption cell on the surface. The adhesion coefficient s_0 was estimated statistically under the assumption of direct nonactivated adsorption [16].

An analysis of the earlier-made calculations [10, 11] showed that on the α - Al_2O_3 surface the oxygen atom adsorption occurs without a barrier, while its mobility on the cluster-modeled cell is small, owing to a fairly high diffusion barrier. Therefore, we have

$$s_0(T) = \frac{q_a^{xy}}{q_0^x q_0^y}.$$

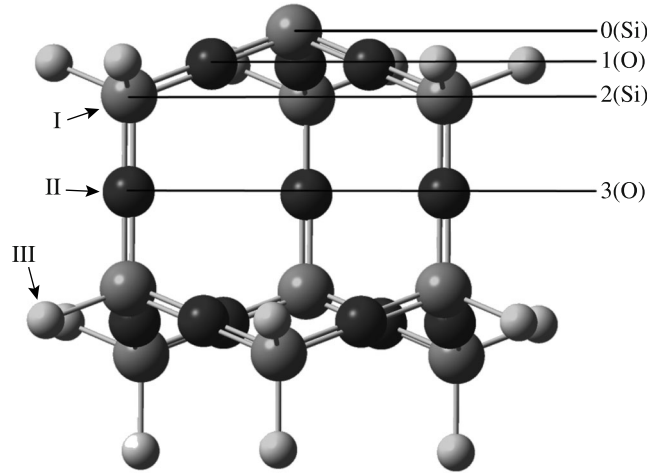


Fig. 3. Structure of the β -cristobalite surface (side view); I are the Si atoms, II are the O atoms, and III are the H atoms.

Here, q_a^{xy} is the statistical sum of the surface oscillations of the adsorption complex and q_0^x and q_0^y are the statistical sums of the two-dimensional translational motion of a free atom on the cell. The Cartesian coordinates x and y correspond to the plane passing through the first-layer atoms of neighboring clusters and z corresponds to the normal to this plane.

The expressions for the statistical sums are as follows:

$$q_0^x q_0^y = \frac{2\pi}{h^2} s_0 m k_B T,$$

$$q_a^{xy} = \exp\left(-\frac{\hbar\omega_x + \hbar\omega_y + \hbar\omega_z}{2k_B T}\right) \sum_{v_x, v_y} \exp\left(-\frac{\hbar\omega_x v_x + \hbar\omega_y v_y}{2k_B T}\right),$$

where ω_α are three vibrational frequencies of an oxygen atom adsorbed on the oxygen atom cluster. The first multiplier allows for the zero oscillation energy, while in the second multiplier the summation is over all states, whose energy does not exceed the adsorption energy ε_{a0} per molecule.

Under the assumption of the local equilibrium between the surface and the gas phase the desorption rate coefficient k_d can be expressed in terms of the equilibrium constant of the adsorption/desorption process

$$K(T) = \frac{k_a}{k_d} = \exp\left(\frac{\varepsilon_{a0}}{k_B T}\right) \frac{q_a(T)}{q_S(T) q_0(T)} N_A V,$$

where q_a , q_S , and q_0 are the total statistical sums of the adsorption complex, the surface, and a free oxygen atom, respectively, N_A is the Avogadro number, and V is the specific volume of the gas phase. In view of the periodic structure of the surface and the explicit expressions for the translational statistical sums of an atom [17], we obtain

$$K(T) = \exp\left(\frac{\varepsilon_{a0}}{k_B T}\right) \left(\frac{2\pi}{h^2} m k_B T\right)^{-1/2} \frac{q_a(T)}{q_S(T) q_0^{el}(T)} \frac{N_A}{S^0},$$

where q_0^{el} is the electronic statistical sum of an atom calculated in the nonrelativistic approximation $^{2S+1}L^\sigma$ (S and L are the values of the electron spin and orbital angular moments) from the reference data [18] with account of the statistical weight $(2S + 1)(2L + 1)$.

Under the assumption that in the interaction with an atom from the gas phase the surface deformation is local, the ratio q_a/q_S can be estimated as the ratio of the statistical sums of the two above-mentioned clusters q_{C-O}/q_C . In these clusters the oscillational components are important; these were estimated by direct summation but with account of the frequencies of all 57 and 54 normal oscillations and a constraint on the states lying below the desorption limit ε_{a0} .

From an analysis of the numerical investigations performed it follows that in the case of the α -Al₂O₃ surface the heterogeneous processes connected with an adsorbed nitrogen atom do not play any considerable role owing to the negligibly small adsorption energy (0.074 eV). The comparison of the calculated equilibrium constants of the absorption of oxygen on α -Al₂O₃ and nitrogen on the β -cristobalite surface showed that for α -Al₂O₃ throughout the entire temperature range under consideration the equilibrium is displaced toward desorption and is controlled by the pre-exponential factor, rather than by the exponential factor itself, while the equilibrium constant is about ten orders smaller than in the case of true chemisorption. Thus, in the case of the α -Al₂O₃ surface the heterogeneous processes connected with an adsorbed nitrogen atom can be completely excluded from consideration.

Impact Recombination (Eley–Rideal Reaction)

The rate coefficients of impact heterogeneous atom recombination on the β -cristobalite surface were calculated on the basis of the molecular- dynamic calculations from the literature [19], where the method of classical trajectories was used.

The rate coefficient of the heterogeneous recombination on the Eley–Rideal surface k_{er} can be represented in the form:

$$k_{er}(T) = \left(\frac{k_B T}{2\pi m} \right)^{1/2} \frac{p(T)}{S^0} N_A.$$

The writing in the Arrhenius form

$$k_{er}(T) = A_{er}(T) \exp \left(-\frac{E_{er}}{k_B T} \right)$$

makes it possible to separate out the pre-exponential factor $A_{er}(T)$.

In [19] on the basis of an investigation of the polymorphic model of the β -cristobalite (100) surface using the density functional method analytical approximations of the pairwise potentials of the nitrogen atom interaction with the surface were derived and used to determine the value of the energy barrier of the Eley–Rideal recombination reaction (0.1 eV) and to perform the molecular-dynamic calculations of the recombination probability. The check showed that the potentials presented in [19] fairly adequately approximate the cluster model data and give a very close value of the activation energy for the β -cristobalite (111) surface. Since in estimating the energy barriers both the approximation accuracy and that of the density functional method are within a few tenths of an electron-volt, in the calculations the energy activation value was taken to be $E_{er} = 0.1$ eV.

In [19] the recombination probabilities $P(v_z)$ are given as functions of the velocity v_z of an atom normally incident on the surface. The temperature dependence of these quantities is determined by the integral transformation with the Maxwellian distribution

$$P(T) = \left(\frac{m}{2\pi k_B T} \right)^{1/2} \int_0^\infty P(v_z) \exp \left(-\frac{mv_z^2}{2k_B T} \right) dv_z,$$

where m is the atom mass and k_B is the Boltzmann constant.

The calculations using the density functional method showed that the activation energies of the Eley–Rideal reaction for nitrogen and oxygen are similar in value. The normalization of the pre-exponential factor by the atom mass makes it possible to obtain a set of the nitrogen and oxygen recombination rate coefficients on the β -cristobalite surface.

The results of the calculations presented in Fig. 4 indicate that the nitrogen recombination is somewhat more effective than the oxygen recombination, while the effective recombination coefficients on the sides of the β -cristobalite (100) and (111) almost coincide at the expense of the Eley–Rideal mechanism. In the figure the rate constant of the oxygen recombination on the (100) surface calculated in [20] on the basis of

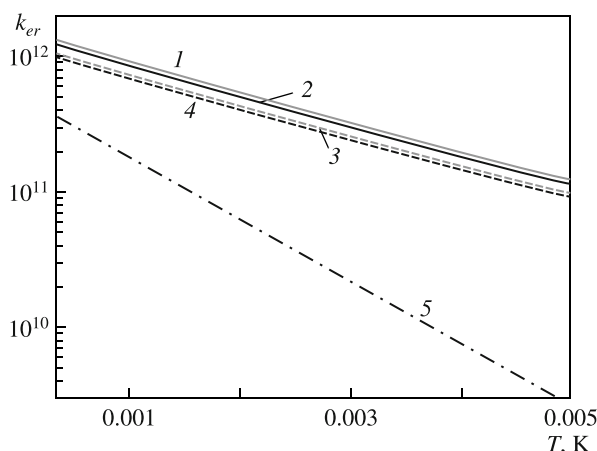


Fig. 4. Rate coefficients [$\text{cm}^3 (\text{s mole})^{-1}$] of heterogeneous oxygen and nitrogen recombination in accordance with the Eley–Rideal mechanism for two β -cristobalite surfaces: (1) N(111); (2) O(111); (3) N(100); (4) O(100); and (5) O(100) from [22].

the periodic model of the surface is also presented. The agreement accurate to the factor of three seems quite acceptable for the kinetic modeling.

On the α - Al_2O_3 surface the oxygen recombination rate in the Eley–Rideal reaction was calculated in [10, 11] using the molecular-dynamic method on the basis of the surface of the potential energy of the interaction between an O atom and the Al_8O_{12} cluster as a function of the distance between the oxygen atom and an $\text{Al}_{(1)}$ atom and two polar angles Θ and φ . In this case, the cluster geometry and the potential curve of the interaction between an O_2 molecule with the Al_4O_6 cluster as a function of the distance from the molecule to the central Al atom were partially optimized. Here, the angle Θ corresponds to the deviation of the $\text{O}-\text{Al}_{(1)}$ vector from the normal to the z surface and the angle φ describes the position of the projection of this vector on the xy plane.

Associative Recombination (Langmuire–Hinshelwood Reaction)

Assuming that the activation energy of recombination of two atoms with simultaneous desorption of a molecule is independent of the way of delivering the reactant to the reaction center the recombination rate coefficient in the Langmuire–Hinshelwood reaction can be written in the form:

$$k_{lh} = \frac{v_{\text{diff}}(T)}{S^0} \exp\left(\frac{E_{er}}{k_B T}\right), \quad v_{\text{diff}}(T) = 4S^0 N_A D(T),$$

where v_{diff} is the frequency of the oxygen atom jumps between the neighboring adsorption sites.

In calculating the diffusion coefficient $D(T)$ we used the empirical scheme in which four clusters are united into a “rhombus” shown in Fig. 5, while the surface of the potential energy of an oxygen atom was approximated by the sum of four surfaces of the potential energy of its interaction with each of these clusters and the Lennard–Jones potentials of its interaction with the $\text{O}_{(2)}$ second-layer oxygen atoms [21]. On this surface the reaction path connecting two nearest $\text{Al}_{(1)}$ atoms with the geometric distance between them $\rho = 4.529 \text{ \AA}$ was optimized and the energy at the maximum point was determined, together with its derivatives. The derivatives correspond to one imaginary and two real frequencies. In view of the symmetry of the problem, there exist two equivalent paths with maxima at the points $(x, y, z) = (2.355, \pm 1.185, 1.579)$. The values of $D(T)$ were calculated within the framework of activated complex theory [22] using the expression

$$D(T) = \rho^2 \frac{k_B T}{2\pi\hbar} \frac{q^\ddagger(T)}{q_q(T)} \exp\left(\frac{E_{D0}}{k_B T}\right),$$

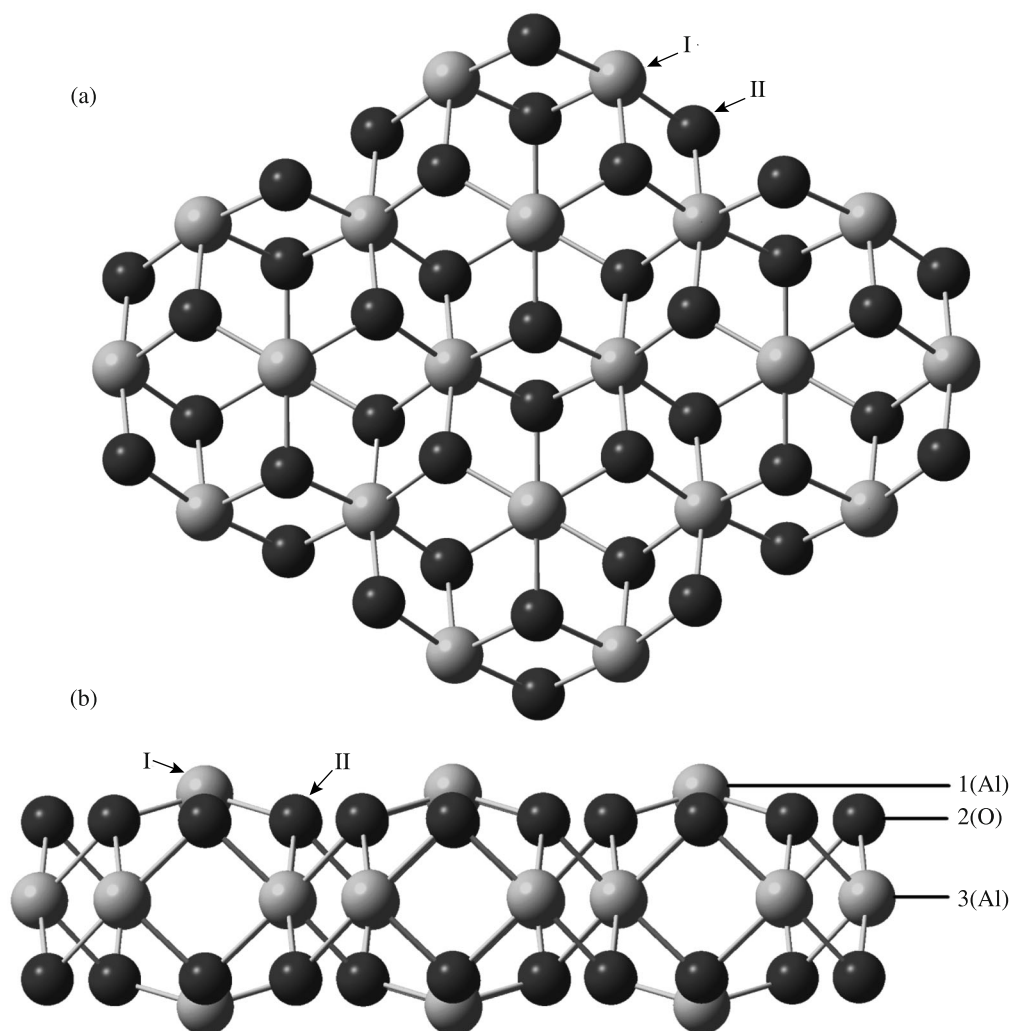


Fig. 5. Model of the α - Al_2O_3 (0001) surface; top (a) and side (b) views on a structural element of the surface obtained by fourfold translation of the Al_8O_{12} cluster; I are the Al atoms and II are the O atoms.

where $q^\ddagger(T)$ is the statistical sum of the transitional state corresponding to a maximum on the reaction path and E_{D0} is the diffusion activation energy with account of the zero oscillation energy per one molecule. Since the model does not take account of surface relaxation in the process of diffusion, in the statistical sum ratio only the contributions connected with the degrees of freedom of an oxygen atom are not cancelled. They include two real vibrational frequencies $w_i^\ddagger$ in the directions orthogonal to the reaction path in the transitional state and three vibrational frequencies ω_α introduced above in the adsorption complex. The corresponding contributions were calculated by direct summation with a constraint on the adsorption energy.

In modeling the associative recombination processes the data [20] were used for determining the corresponding reaction rate for oxygen recombination on the β -cristobalite surface. Their approximation on the temperature range from 700 to 1700 K gave the values of the pre-exponential factor $1.0 \times 10^8 \text{ cm}^2/(\text{s mole})$ and the reaction activation energy 4.24 eV. The rate coefficient of the associative atom recombination was calculated using an algorithm of extrapolation in the mass under the assumption that for a particular surface the diffusion activation energy is determined by the difference of the adsorption energies of atomic components.

The recommended values of the rate constants of the elementary stages of the interaction between dissociated air and thermal-protective β -cristobalite and α - Al_2O_3 coatings are gathered in the table.

Table

Rate coefficient	β -cristobalite, $k = A \exp(-E/RT)$		α -Al ₂ O ₃ , $k = A \exp(-E/RT)$	
	A	E, eV	A	E, eV
k_{aN} , cm ³ (s mole) ⁻¹	$1.57 \times 10^8 T^{1.1}$	0.0	0	0
k_{dN} , s ⁻¹	6.0×10^{10}	3.472	0	0
k_{aO} , cm ³ (s mole) ⁻¹	$1.05 \times 10^8 T^{1.1}$	0.0	$3.778 \times 10^9 T$	-0.039
k_{dO} , s ⁻¹	2.0×10^{11}	5.051	4.826×10^{11}	1.666
k_{erN} , cm ³ (s mole) ⁻¹	1.6×10^{12}	0.047	0	0
k_{erO} , cm ³ (s mole) ⁻¹	1.48×10^{12}	0.046	$3.1061 \times 10^{12} \sqrt{T}$	0.43
k_{lhO} , cm ² s ⁻¹	1.0×10^8	4.24	1.614×10^{21}	2.51
k_{lhN} , cm ² s ⁻¹	1.1×10^8	2.7	0	0

Summary. It is shown that under the re-entry conditions the α -Al₂O₃ material is a considerably weaker adsorbate of oxygen and nitrogen atoms, as compared with β -cristobalite. Its surface does not almost keep nitrogen atoms and weakly adsorbs oxygen atoms. The difference in the atom adsorption energies on the α -Al₂O₃ and β -cristobalite surfaces is greater than 3 eV and is almost independent (within the limits of 0.1 eV) of the adsorbed atom nature.

On the basis of the quantum-chemical investigations and an analysis and processing of the experimental and calculated data available in the literature we can recommend the given values of the reaction rates of the elementary stages of the process of interaction between dissociated air and thermal-protective α -Al₂O₃ and β -cristobalite coatings.

We note that the energy of oxygen atom desorption from the β -cristobalite surface determined in the quantum-mechanical calculations is considerably greater than the data available in the literature. By halving this value good agreement with the experimental data can be obtained.

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